Unit 2

What is Data Collection?

Data collection is the process of collecting, measuring and analysing different types of information using a set of standard validated techniques. The main objective of data collection is to gather information-rich and reliable data, and analyse them to make critical business decisions. Once the data is collected, it goes through a rigorous process of data cleaning and data processing to make this data truly useful for businesses.

There are two main methods of data collection in research based on the information that is required, namely:

- Primary Data Collection
- Secondary Data Collection

Primary Data Collection Methods

- Primary data refers to data collected from first-hand experience directly from the main source. It refers to data that has never been used in the past. The data gathered by primary data collection methods are generally regarded as the best kind of data in research.
- The methods of collecting primary data can be further divided into quantitative data collection methods (deals with factors that can be counted) and qualitative data collection methods (deals with factors that are not necessarily numerical in nature).
- **Here are some of the most common primary data collection methods:**

1. Interviews

Interviews are a direct method of data collection. It is simply a process in which the interviewer asks questions and the interviewee responds to them. It provides a high degree of flexibility because questions can be adjusted and changed anytime according to the situation.

2. Observations

In this method, researchers observe a situation around them and record the findings. It can be used to evaluate the behaviour of different people in controlled (everyone knows they are being observed) and uncontrolled (no one knows they are being observed) situations. This method is highly effective because it is straightforward and not directly dependent on other participants.

For example, a person looks at random people that walk their pets on a busy street, and then uses this data to decide whether or not to open a pet food store in that area.

3. Surveys and Questionnaires

Surveys and questionnaires provide a broad perspective from large groups of people. They can be conducted face-to-face, mailed, or even posted on the Internet to get respondents from anywhere in the world. The answers can be yes or no, true or false, multiple choice, and even open-ended questions. However, a drawback of surveys and questionnaires is delayed response and the possibility of ambiguous answers.

4. Focus Groups

A focus group is similar to an interview, but it is conducted with a group of people who all have something in common. The data collected is similar to in-person interviews, but they offer a better understanding of why a certain group of people thinks in a particular way. However, some drawbacks of this method are lack of privacy and domination of the interview by one or two participants. Focus groups can also be time-consuming and challenging, but they help reveal some of the best information for complex situations.

5. Projective Data Gathering

Projective data gathering is an indirect interview, used when potential respondents know why they're being asked questions and hesitate to answer. For instance, someone may be reluctant to answer questions about their phone service if a cell phone carrier representative poses the questions. With projective data gathering, the interviewees get an incomplete question, and they must fill in the rest, using their opinions, feelings, and attitudes.

6. **Delphi Technique**
   The Oracle at Delphi, according to Greek mythology, was the high priestess of Apollo's temple, who gave advice, prophecies, and counsel. In the realm of data collection, researchers use the Delphi technique by gathering information from a panel of experts. Each expert answers questions in their field of specialty, and the replies are consolidated into a single opinion.
7. **Questionnaires**
   Questionnaires are a simple, straightforward data collection method. Respondents get a series of questions, either open or close-ended, related to the matter at hand.

5. Oral Histories

Oral histories also involve asking questions like interviews and focus groups. However, it is defined more precisely and the data collected is linked to a single phenomenon. It involves collecting the opinions and personal experiences of people in a particular event that they were involved in. For example, it can help in studying the effect of a new product in a particular community.

Secondary Data Collection Methods

Secondary data refers to data that has already been collected by someone else. It is much more inexpensive and easier to collect than primary data. While primary data collection provides more authentic and original data, there are numerous instances where secondary data collection provides great value to organizations.

Here are some of the most common secondary data collection methods:

1. Internet

The use of the Internet has become one of the most popular secondary data collection methods in recent times. There is a large pool of free and paid research resources that can be easily accessed on the Internet. While this method is a fast and easy way of data collection, you should only source from authentic sites while collecting information.

2. Government Archives

There is lots of data available from government archives that you can make use of. The most important advantage is that the data in government archives are authentic and verifiable. The challenge, however, is that data is not always readily available due to a number of factors. For example, criminal records can come under classified information and are difficult for anyone to have access to them.

3. Libraries

Most researchers donate several copies of their academic research to libraries. You can collect important and authentic information based on different research contexts. Libraries also serve as a storehouse for business directories, annual reports and other similar documents that help businesses in their research.

Unlike primary data collection, there are no specific collection methods. Instead, since the information has already been collected, the researcher consults various data sources, such as:

4. **Financial Statements**
5. **Sales Reports**
6. **Retailer/Distributor/Deal Feedback**
7. **Customer Personal Information (e.g., name, address, age, contact info)**
8. **Business Journals**

9. **Government Records (e.g., census, tax records, Social Security info)**
10. **Trade/Business Magazines**
11. **The internet**

Data Collection Tools

Now that we've explained the various techniques, let's narrow our focus even further by looking at some specific tools. For example, we mentioned interviews as a technique, but we can further break that down into different interview types (or "tools").

Word Association
The researcher gives the respondent a set of words and asks them what comes to mind when they hear each word.

Sentence Completion
Researchers use sentence completion to understand what kind of ideas the respondent has. This tool involves giving an incomplete sentence and seeing how the interviewee finishes it.

Role-Playing
Respondents are presented with an imaginary situation and asked how they would act or react if it was real.

In-Person Surveys
The researcher asks questions in person.

Online/Web Surveys
These surveys are easy to accomplish, but some users may be unwilling to answer truthfully, if at all.

Mobile Surveys
These surveys take advantage of the increasing proliferation of mobile technology. Mobile collection surveys rely on mobile devices like tablets or smartphones to conduct surveys via SMS or mobile apps.

Phone Surveys
No researcher can call thousands of people at once, so they need a third party to handle the chore. However, many people have call screening and won't answer.

Observation
Sometimes, the simplest method is the best. Researchers who make direct observations collect data quickly and easily, with little intrusion or third-party bias. Naturally, it's only effective in small-scale situations.
What are Common Challenges in Data Collection?
There are some prevalent challenges faced while collecting data, let us explore a few of them to understand them better and avoid them.

Data Quality Issues
The main threat to the broad and successful application of machine learning is poor data quality. Data quality must be your top priority if you want to make technologies like machine learning work for you. Let's talk about some of the most prevalent data quality problems in this blog article and how to fix them.

Inconsistent Data
When working with various data sources, it's conceivable that the same information will have discrepancies between sources. The differences could be in formats, units, or occasionally spellings. The introduction of inconsistent data might also occur during firm mergers or relocations. Inconsistencies in data have a tendency to accumulate and reduce the value of data if they are not continually resolved. Organizations that have heavily focused on data consistency do so because they only want reliable data to support their analytics.

Data Downtime
Data is the driving force behind the decisions and operations of data-driven businesses. However, there may be brief periods when their data is unreliable or not prepared. Customer complaints and subpar analytical outcomes are only two ways that this data unavailability can have a significant impact on businesses. A data engineer spends about 80% of their time updating, maintaining, and guaranteeing the integrity of the data pipeline. In order to ask the next business question, there is a high marginal cost due to the lengthy operational lead time from data capture to insight.

Schema modifications and migration problems are just two examples of the causes of data downtime. Data pipelines can be difficult due to their size and complexity. Data downtime must be continuously monitored, and it must be reduced through automation.

Ambiguous Data
Even with thorough oversight, some errors can still occur in massive databases or data lakes. For data streaming at a fast speed, the issue becomes more overwhelming. Spelling mistakes can go unnoticed, formatting difficulties can occur, and column heads might be deceptive. This unclear data might cause a number of problems for reporting and analytics.
Duplicate Data
Streaming data, local databases, and cloud data lakes are just a few of the sources of data that modern enterprises must contend with. They might also have application and system silos. These sources are likely to duplicate and overlap each other quite a bit. For instance, duplicate contact information has a substantial impact on customer experience. If certain prospects are ignored while others are engaged repeatedly, marketing campaigns suffer. The likelihood of biased analytical outcomes increases when duplicate data are present. It can also result in ML models with biased training data.

Too Much Data
While we emphasize data-driven analytics and its advantages, a data quality problem with excessive data exists. There is a risk of getting lost in an abundance of data when searching for information pertinent to your analytical efforts. Data scientists, data analysts, and business users devote 80% of their work to finding and organizing the appropriate data. With an increase in data volume, other problems with data quality become more serious, particularly when dealing with streaming data and big files or databases.

Inaccurate Data
For highly regulated businesses like healthcare, data accuracy is crucial. Given the current experience, it is more important than ever to increase the data quality for COVID-19 and later pandemics. Inaccurate information does not provide you with a true picture of the situation and cannot be used to plan the best course of action. Personalized customer experiences and marketing strategies underperform if your customer data is inaccurate.
Hidden Data
The majority of businesses only utilize a portion of their data, with the remainder sometimes being lost in data silos or discarded in data graveyards. For instance, the customer service team might not

receive client data from sales, missing an opportunity to build more precise and comprehensive customer profiles. Missing out on possibilities to develop novel products, enhance services, and streamline procedures is caused by hidden data.

Finding Relevant Data
Finding relevant data is not so easy. There are several factors that we need to consider while trying to find relevant data, which include -

Relevant Domain
Relevant demographics
Relevant Time period and so many more factors that we need to consider while trying to find relevant data.
Data that is not relevant to our study in any of the factors render it obsolete and we cannot effectively proceed with its analysis. This could lead to incomplete research or analysis, re-collecting data again and again, or shutting down the study.

Deciding the Data to Collect
Determining what data to collect is one of the most important factors while collecting data and should be one of the first factors while collecting data. We must choose the subjects the data will cover, the sources we will be used to gather it, and the quantity of information we will require. Our responses to these queries will depend on our aims, or what we expect to achieve utilizing your data. As an illustration, we may choose to gather information on the categories of articles that website visitors between the ages of 20 and 50 most frequently access. We can also decide to compile data on the typical age of all the clients who made a purchase from your business over the previous month.

Not addressing this could lead to double work and collection of irrelevant data or ruining your study as a whole.

Dealing With Big Data
Big data refers to exceedingly massive data sets with more intricate and diversified structures. These traits typically result in increased challenges while storing, analyzing, and using additional methods of extracting results. Big data refers especially to data sets that are quite enormous or intricate that conventional data processing tools are insufficient. The overwhelming amount of data, both unstructured and structured, that a business faces on a daily basis.

The amount of data produced by healthcare applications, the internet, social networking sites social, sensor networks, and many other businesses are rapidly growing as a result of recent technological advancements. Big data refers to the vast volume of data created from numerous sources in a variety of formats at extremely fast rates. Dealing with this kind of data is one of the many challenges of Data Collection and is a crucial step toward collecting effective data.

Low Response and Other Research Issues

Poor design and low response rates were shown to be two issues with data collecting, particularly in health surveys that used questionnaires. This might lead to an insufficient or inadequate supply of data for the study. Creating an incentivized data collection program might be beneficial in this case to get more responses.

Data Cleaning:- Data cleaning is one of the important parts of machine learning. It plays a significant part in building a model. It surely isn't the fanciest part of machine learning and at the same time, there aren't any hidden tricks or secrets to uncover. However, the success or failure of a project relies on proper data cleaning. Professional data scientists usually invest a very large portion of their time in this step because of the belief that "Better data beats fancier algorithms".

If we have a well-cleaned dataset, there are chances that we can get achieve good results with simple algorithms also, which can prove very beneficial at times especially in terms of computation when the dataset size is large. Obviously, different types of data will require different types of cleaning. However, this systematic approach can always serve as a good starting point.

Steps of Data Cleaning

While the techniques used for data cleaning may vary according to the types of data your company stores, you can follow these basic steps to cleaning your data, such as:

1. Remove duplicate or irrelevant observations

Remove unwanted observations from your dataset, including duplicate observations or irrelevant observations. Duplicate observations will happen most often during data collection. When you combine data sets from multiple places, scrape data, or receive data from clients or multiple departments, there are opportunities to create duplicate data. De-duplication is one of the largest areas to be considered in this process. Irrelevant observations are when you notice observations that do not fit into the specific problem you are trying to analyze.

For example, if you want to analyze data regarding millennial customers, but your dataset includes older generations, you might remove those irrelevant observations. This can make analysis more efficient, minimize distraction from your primary target, and create a more manageable and performable dataset.

2. Fix structural errors

Structural errors are when you measure or transfer data and notice strange naming conventions, typos, or incorrect capitalization. These inconsistencies can cause mislabeled categories or classes. For example, you may find "N/A" and "Not Applicable" in any sheet, but they should be analyzed in the same category.

3. Filter unwanted outliers

Often, there will be one-off observations where, at a glance, they do not appear to fit within the data you are analyzing. If you have a legitimate reason to remove an outlier, like improper data entry, doing so will help the performance of the data you are working with.

However, sometimes, the appearance of an outlier will prove a theory you are working on. And just because an outlier exists doesn't mean it is incorrect. This step is needed to determine the validity of that number. If an outlier proves to be irrelevant for analysis or is a mistake, consider removing it.

4. Handle missing data

You can't ignore missing data because many algorithms will not accept missing values. There are a couple of ways to deal with missing data. Neither is optimal, but both can be considered, such as:

- You can drop observations with missing values, but this will drop or lose information, so be careful before removing it.
- You can input missing values based on other observations; again, there is an opportunity to lose the integrity of the data because you may be operating from assumptions and not actual observations.
- You might alter how the data is used to navigate null values effectively.

5. Validate and QA

At the end of the data cleaning process, you should be able to answer these questions as a part of basic validation, such as:

- Does the data make sense?
- Does the data follow the appropriate rules for its field?
- Does it prove or disprove your working theory or bring any insight to light?
- Can you find trends in the data to help you for your next theory?
- If not, is that because of a data quality issue?

Because of incorrect or noisy data, false conclusions can inform poor business strategy and decision-making. False conclusions can lead to an embarrassing moment in a reporting meeting when you realize your data doesn't stand up to study. Before you get there, it is important to create a culture of quality data in your organization. To do this, you should document the tools you might use to create this strategy.

Methods of Data Cleaning

There are many data cleaning methods through which the data should be run. The methods are described below:

1. **Ignore the tuples:** This method is not very feasible, as it only comes to use when the tuple has several attributes is has missing values.
2. **Fill the missing value:** This approach is also not very effective or feasible. Moreover, it can be a time-consuming method. In the approach, one has to fill in the missing value. This is usually done manually, but it can also be done by attribute mean or using the most probable value.
3. **Binning method:** This approach is very simple to understand. The smoothing of sorted data is done using the values around it. The data is then divided into several segments of equal size. After that, the different methods are executed to complete the task.
4. **Regression:** The data is made smooth with the help of using the regression function. The regression can be linear or multiple. Linear regression has only one independent variable, and multiple regressions have more than one independent variable.
5. **Clustering:** This method mainly operates on the group. Clustering groups the data in a cluster. Then, the outliers are detected with the help of clustering. Next, the similar values are then arranged into a "group" or a "cluster".

Process of Data Cleaning

The following steps show the process of data cleaning in data mining.

1. **Monitoring the errors:** Keep a note of suitability where the most mistakes arise. It will make it easier to determine and stabilize false or corrupt information. Information is especially necessary while integrating another possible alternative with established management software.
2. **Standardize the mining process:** Standardize the point of insertion to assist and reduce the chances of duplicity.
3. **Validate data accuracy:** Analyze and invest in data tools to clean the record in real-time. Tools used Artificial Intelligence to better examine for correctness.
4. **Scrub for duplicate data:** Determine duplicates to save time when analyzing data. Frequently attempted the same data can be avoided by analyzing and investing in separate data erasing tools that can analyze rough data in quantity and automate the operation.
5. **Research on data:** Before this activity, our data must be standardized, validated, and scrubbed for duplicates. There are many third-party sources, and these Approved & authorized parties sources can capture information directly from our databases. They help us to clean and compile the data to ensure completeness, accuracy, and reliability for business decision-making.
6. **Communicate with the team:** Keeping the group in the loop will assist in developing and strengthening the client and sending more targeted data to prospective customers.

Tools for Data Cleaning in Data Mining

Data Cleansing Tools can be very helpful if you are not confident of cleaning the data yourself or have no time to clean up all your data sets. You might need to invest in those tools, but it is worth the expenditure. There are many data cleaning tools in the market. Here are some top-ranked data cleaning tools, such as:

1. OpenRefine
2. Trifacta Wrangler
3. Drake
4. Data Ladder
5. Data Cleaner
6. Cloudingo
7. Reifier
8. IBM Infosphere Quality Stage
9. TIBCO Clarity
10. Winpure

Benefits of Data Cleaning

Having clean data will ultimately increase overall productivity and allow for the highest quality information in your decision-making. Here are some major benefits of data cleaning in data mining, such as:

- Removal of errors when multiple sources of data are at play.
- Fewer errors make for happier clients and less-frustrated employees.
- Ability to map the different functions and what your data is intended to do.
- Monitoring errors and better reporting to see where errors are coming from, making it easier to fix incorrect or corrupt data for future applications.
- Using tools for data cleaning will make for more efficient business practices and quicker decision-making.

Data Integration:- Data integration in data mining refers to the process of combining data from multiple sources into a single, unified view. This can involve cleaning and transforming the data, as well as resolving any inconsistencies or conflicts that may exist between the different sources. The goal of data integration is to make the data more useful and meaningful for the purposes of analysis and decision making. Techniques used in data integration include data warehousing, ETL (extract, transform, load) processes, and data federation. Data Integration is a data pre-processing technique that combines data from multiple heterogeneous data sources into a coherent data store and provides a unified view of the data. These sources may include multiple data cubes, databases, or flat files.

The data integration approaches are formally defined as triple <G, S, M> where,

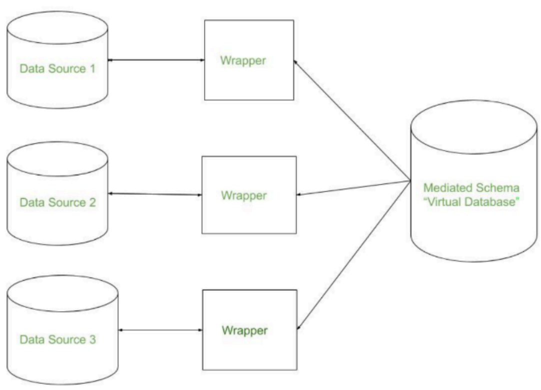
G stand for the global schema,

S stands for the heterogeneous source of schema,

M stands for mapping between the queries of source and global schema.

What is data integration: Data integration is the process of combining data from multiple sources into a cohesive and consistent view. This process involves identifying and accessing the different data sources, mapping the data to a common format, and reconciling any inconsistencies or discrepancies between the sources. The goal of data integration is to make it easier to access and analyze data that is spread across multiple systems or platforms, in order to gain a more complete and accurate understanding of the data.

Data integration is used in a wide range of applications, such as business intelligence, data warehousing, master data management, and analytics. Data integration can be critical to the success of these applications, as it enables organizations to access and analyze data that is spread across different systems, departments, and lines of business, in order to make better decisions, improve operational efficiency, and gain a competitive advantage.



There are mainly 2 major approaches for data integration – one is the "tight coupling approach" and another is the "loose coupling approach".

Tight Coupling:
This approach involves creating a centralized repository or data warehouse to store the integrated data. The data is extracted from various sources, transformed and loaded into a data warehouse. Data

is integrated in a tightly coupled manner, meaning that the data is integrated at a high level, such as at the level of the entire dataset or schema. This approach is also known as data warehousing, and it enables data consistency and integrity, but it can be inflexible and difficult to change or update.

- Here, a data warehouse is treated as an information retrieval component.
- In this coupling, data is combined from different sources into a single physical location through the process of ETL – Extraction, Transformation, and Loading.

Loose Coupling:

This approach involves integrating data at the lowest level, such as at the level of individual data elements or records. Data is integrated in a loosely coupled manner, meaning that the data is integrated at a low level, and it allows data to be integrated without having to create a central repository or data warehouse. This approach is also known as data federation, and it enables data flexibility and easy updates, but it can be difficult to maintain consistency and integrity across multiple data sources.

- Here, an interface is provided that takes the query from the user, transforms it in a way the source database can understand, and then sends the query directly to the source databases to obtain the result.
- And the data only remains in the actual source databases.

Issues in Data Integration:

There are several issues that can arise when integrating data from multiple sources, including:

1. **Data Quality:** Inconsistencies and errors in the data can make it difficult to combine and analyze.
2. **Data Semantics:** Different sources may use different terms or definitions for the same data, making it difficult to combine and understand the data.
3. **Data Heterogeneity:** Different sources may use different data formats, structures, or schemas, making it difficult to combine and analyze the data.
4. **Data Privacy and Security:** Protecting sensitive information and maintaining security can be difficult when integrating data from multiple sources.
5. **Scalability:** Integrating large amounts of data from multiple sources can be computationally expensive and time-consuming.
6. **Data Governance:** Managing and maintaining the integration of data from multiple sources can be difficult, especially when it comes to ensuring data accuracy, consistency, and timeliness.
7. **Performance:** Integrating data from multiple sources can also affect the performance of the system.
8. **Integration with existing systems:** Integrating new data sources with existing systems can be a complex task, requiring significant effort and resources.
9. **Complexity:** The complexity of integrating data from multiple sources can be high, requiring specialized skills and knowledge.

There are three issues to consider during data integration: Schema Integration, Redundancy Detection, and resolution of data value conflicts. These are explained in brief below.

1. Schema Integration:

- Integrate metadata from different sources.
- The real-world entities from multiple sources are referred to as the entity identification problem.ER

2. Redundancy Detection:

- An attribute may be redundant if it can be derived or obtained from another attribute or set of attributes.
- Inconsistencies in attributes can also cause redundancies in the resulting data set.
- Some redundancies can be detected by correlation analysis.

3. Resolution of data value **conflicts:**

- This is the third critical issue in data integration.
- Attribute values from different sources may differ for the same real-world entity.
- An attribute in one system may be recorded at a lower level of abstraction than the “same” attribute in another.

Data Transformation:-

Data transformation in data mining refers to the process of converting raw data into a format that is suitable for analysis and modeling. The goal of data transformation is to prepare the data for data mining so that it can be used to extract useful insights and knowledge. Data transformation typically involves several steps, including:

1. **Data cleaning:** Removing or correcting errors, inconsistencies, and missing values in the data.
2. **Data integration:** Combining data from multiple sources, such as databases and spreadsheets, into a single format.
3. **Data normalization:** Scaling the data to a common range of values, such as between 0 and 1, to facilitate comparison and analysis.
4. **Data reduction:** Reducing the dimensionality of the data by selecting a subset of relevant features or attributes.
5. **Data discretization**: Converting continuous data into discrete categories or bins.
6. **Data aggregation:** Combining data at different levels of granularity, such as by summing or averaging, to create new features or attributes.
7. Data transformation is an important step in the data mining process as it helps to ensure that the data is in a format that is suitable for analysis and modeling, and that it is free of errors and inconsistencies. Data transformation can also help to improve the performance of data mining algorithms, by reducing the dimensionality of the data, and by scaling the data to a common range of values.

The data are transformed in ways that are ideal for mining the data. The data transformation involves steps that are:

1. Smoothing: It is a process that is used to remove noise from the dataset using some algorithms It allows for highlighting important features present in the dataset. It helps in predicting the patterns. When collecting data, it can be manipulated to eliminate or reduce any variance or any other noise form. The concept behind data smoothing is that it will be able to identify simple changes to help predict different trends and patterns. This serves as a help to analysts or traders who need to look at a lot of data which can often be difficult to digest for finding patterns that they wouldn't see otherwise.
2. Aggregation: Data collection or aggregation is the method of storing and presenting data in a summary format. The data may be obtained from multiple data sources to integrate these data sources into a data analysis description. This is a crucial step since the accuracy of data analysis insights is highly dependent on the quantity and quality of the data used. Gathering accurate data of high quality and a large enough quantity is necessary to produce relevant results. The collection of data is useful for everything from decisions concerning financing or business strategy of the product, pricing, operations, and marketing strategies. For **example**, Sales, data may be aggregated to compute monthly& annual total amounts.
3. Discretization: It is a process of transforming continuous data into set of small intervals. Most Data Mining activities in the real world require continuous attributes. Yet many of the existing data mining frameworks are unable to handle these attributes. Also, even if a data mining task can manage a continuous attribute, it can significantly improve its efficiency by replacing a constant quality attribute with its discrete values. For **example**, (1-10, 11-20) (age:- young, middle age, senior).
4. Attribute Construction: Where new attributes are created & applied to assist the mining process from the given set of attributes. This simplifies the original data & makes the mining more efficient.
5. Generalization: It converts low-level data attributes to high-level data attributes using concept hierarchy. For Example Age initially in Numerical form (22, 25) is converted into categorical value (young, old). For **example**, Categorical attributes, such as house addresses, may be generalized to higher-level definitions, such as town or country.
6. Normalization: Data normalization involves converting all data variables into a given range. Techniques that are used for normalization are:

- **Min-Max Normalization:**
    - This transforms the original data linearly.
    - Suppose that: min_A is the minima and max_A is the maxima of an attribute, P

    - Where v is the value you want to plot in the new range.
    - v' is the new value you get after normalizing the old value.
- **Z-Score Normalization:**
    - In z-score normalization (or zero-mean normalization) the values of an attribute (A), are normalized based on the mean of A and its standard deviation
    - A value, v, of attribute A is normalized to v' by computing
- **Decimal Scaling:**
    - It normalizes the values of an attribute by changing the position of their decimal points
    - The number of points by which the decimal point is moved can be determined by the absolute maximum value of attribute A.
    - A value, v, of attribute A is normalized to v' by computing
    - where j is the smallest integer such that Max(|v'|) < 1.
    - Suppose: Values of an attribute P varies from -99 to 99.
    - The maximum absolute value of P is 99.
    - For normalizing the values we divide the numbers by 100 (i.e., j = 2) or (number of integers in the largest number) so that values come out to be as 0.98, 0.97 and so on.

ADVANTAGES OR DISADVANTAGES:

Advantages of Data Transformation in Data Mining:

1. Improves Data Quality: Data transformation helps to improve the quality of data by removing errors, inconsistencies, and missing values.
2. Facilitates Data Integration: Data transformation enables the integration of data from multiple sources, which can improve the accuracy and completeness of the data.
3. Improves Data Analysis: Data transformation helps to prepare the data for analysis and modeling by normalizing, reducing dimensionality, and discretizing the data.
4. Increases Data Security: Data transformation can be used to mask sensitive data, or to remove sensitive information from the data, which can help to increase data security.
5. Enhances Data Mining Algorithm Performance: Data transformation can improve the performance of data mining algorithms by reducing the dimensionality of the data and scaling the data to a common range of values.

Disadvantages of Data Transformation in Data Mining:

1. Time-consuming: Data transformation can be a time-consuming process, especially when dealing with large datasets.
2. Complexity: Data transformation can be a complex process, requiring specialized skills and knowledge to implement and interpret the results.
3. Data Loss: Data transformation can result in data loss, such as when discretizing continuous data, or when removing attributes or features from the data.
4. Biased transformation: Data transformation can result in bias, if the data is not properly understood or used.
5. High cost: Data transformation can be an expensive process, requiring significant investments in hardware, software, and personnel.

Overfitting: Data transformation can lead to overfitting, which is a common problem in machine learning where a model learns the detail and noise in the training data to the extent that it negatively impacts the performance of the model on new unseen data.

Data Reduction: - Data reduction is a technique used in data mining to reduce the size of a dataset while still preserving the most important information. This can be beneficial in situations where the dataset is too large to be processed efficiently, or where the dataset contains a large amount of irrelevant or redundant information.

There are several different data reduction techniques that can be used in data mining, including:

1. **Data Sampling:** This technique involves selecting a subset of the data to work with, rather than using the entire dataset. This can be useful for reducing the size of a dataset while still preserving the overall trends and patterns in the data.
2. **Dimensionality Reduction:** This technique involves reducing the number of features in the dataset, either by removing features that are not relevant or by combining multiple features into a single feature.
3. **Data Compression:** This technique involves using techniques such as lossy or lossless compression to reduce the size of a dataset.
4. **Data Discretization:** This technique involves converting continuous data into discrete data by partitioning the range of possible values into intervals or bins.
5. **Feature Selection:** This technique involves selecting a subset of features from the dataset that are most relevant to the task at hand.
6. It's important to note that data reduction can have a trade-off between the accuracy and the size of the data. The more data is reduced, the less accurate the model will be and the less generalizable it will be.

In conclusion, data reduction is an important step in data mining, as it can help to improve the efficiency and performance of machine learning algorithms by reducing the size of the dataset. However, it is important to be aware of the trade-off between the size and accuracy of the data, and carefully assess the risks and benefits before implementing it.

Methods of data reduction:

These are explained as following below.

1. **Data Cube Aggregation:**
   This technique is used to aggregate data in a simpler form. For example, imagine the information you gathered for your analysis for the years 2012 to 2014, that data includes the revenue of your company every three months. They involve you in the annual sales, rather than the quarterly average, So we can summarize the data in such a way that the resulting data summarizes the total sales per year instead of per quarter. It summarizes the data.
2. **2. Dimension reduction:**
   Whenever we come across any data which is weakly important, then we use the attribute required for our analysis. It reduces data size as it eliminates outdated or redundant features.

- **Step-wise Forward Selection –**
  The selection begins with an empty set of attributes later on we decide the best of the original attributes on the set based on their relevance to other attributes. We know it as a p-value in statistics.
  Suppose there are the following attributes in the data set in which few attributes are redundant.

Initial attribute Set: {X1, X2, X3, X4, X5, X6}

Initial reduced attribute set: { }

Step-1: {X1}

Step-2: {X1, X2}

Step-3: {X1, X2, X5}

Final reduced attribute set: {X1, X2, X5}

- **Step-wise Backward Selection –**
  This selection starts with a set of complete attributes in the original data and at each point, it eliminates the worst remaining attribute in the set.
  Suppose there are the following attributes in the data set in which few attributes are redundant.

Initial attribute Set: {X1, X2, X3, X4, X5, X6}

Initial reduced attribute set: {X1, X2, X3, X4, X5, X6 }

Step-1: {X1, X2, X3, X4, X5}

Step-2: {X1, X2, X3, X5}

Step-3: {X1, X2, X5}

Final reduced attribute set: {X1, X2, X5}

- **Combination of forwarding and Backward Selection –**
  It allows us to remove the worst and select the best attributes, saving time and making the process faster.

3. Data Compression:
The data compression technique reduces the size of the files using different encoding mechanisms (Huffman Encoding & run-length Encoding). We can divide it into two types based on their compression techniques.

- **Lossless Compression –**
  Encoding techniques (Run Length Encoding) allow a simple and minimal data size reduction. Lossless data compression uses algorithms to restore the precise original data from the compressed data.
- **Lossy Compression –**
  Methods such as the Discrete Wavelet transform technique, PCA (principal component analysis) are examples of this compression. For e.g., the JPEG image format is a lossy compression, but we can find the meaning equivalent to the original image. In lossy-data compression, the decompressed data may differ from the original data but are useful enough to retrieve information from them.

4. Numerosity Reduction:
In this reduction technique, the actual data is replaced with mathematical models or smaller representations of the data instead of actual data, it is important to only store the model parameter. Or non-parametric methods such as clustering, histogram, and sampling.

5. Discretization & Concept Hierarchy Operation:
Techniques of data discretization are used to divide the attributes of the continuous nature into data with intervals. We replace many constant values of the attributes by labels of small intervals. This means that mining results are shown in a concise, and easily understandable way.

- **Top-down discretization –**
  If you first consider one or a couple of points (so-called breakpoints or split points) to divide the whole set of attributes and repeat this method up to the end, then the process is known as top-down discretization also known as splitting.
- **Bottom-up discretization –**
  If you first consider all the constant values as split points, some are discarded through a combination of the neighbourhood values in the interval, that process is called bottom-up discretization.

Advantages:

1. Improved efficiency: Data reduction can help to improve the efficiency of machine learning algorithms by reducing the size of the dataset. This can make it faster and more practical to work with large datasets.
2. Improved performance: Data reduction can help to improve the performance of machine learning algorithms by removing irrelevant or redundant information from the dataset. This can help to make the model more accurate and robust.
3. Reduced storage costs: Data reduction can help to reduce the storage costs associated with large datasets by reducing the size of the data.
4. Improved interpretability: Data reduction can help to improve the interpretability of the results by removing irrelevant or redundant information from the dataset.

Disadvantages:

1. Loss of information: Data reduction can result in a loss of information, if important data is removed during the reduction process.
2. Impact on accuracy: Data reduction can impact the accuracy of a model, as reducing the size of the dataset can also remove important information that is needed for accurate predictions.
3. Impact on interpretability: Data reduction can make it harder to interpret the results, as removing irrelevant or redundant information can also remove context that is needed to understand the results.
4. Additional computational costs: Data reduction can add additional computational costs to the data mining process, as it requires additional processing time to reduce the data.
5. In conclusion, data reduction can have both advantages and disadvantages. It can improve the efficiency and performance of machine learning algorithms by reducing the size of the dataset. However, it can also result in a loss of information, and make it harder to interpret the results. It's important to weigh the pros and cons of data reduction and carefully assess the risks and benefits before implementing it.

What is Data Reduction? Data mining is applied to the selected data in a large amount database. When data analysis and mining is done on a huge amount of data then it takes a very long time to process, which makes it impractical and infeasible. It can reduce the processing time for data analysis, data reduction techniques are used to obtain a reduced representation of the dataset that is much smaller in volume by maintaining the integrity of the original data. By reducing the data, the efficiency of the data mining process is improved which produces the same analytical results.

Data reduction aims to define it more compactly. When the data size is smaller, it is simpler to apply sophisticated and computationally high-priced algorithms. The reduction of the data may be in terms of the number of rows (records) or terms of the number of columns (dimensions).

There are various strategies for data reduction which are as follows −

Data cube aggregation − In this method, where aggregation operations are used to the data in the construction of a data cube. These data include the All Electronics sales per quarter, for the years 2002 to 2004. It is interested in the annual sales (total per year), rather than the total per quarter. Thus the data can be aggregated so that the resulting data summarize the total sales per year instead of per quarter. The resulting data set is smaller in volume, without loss of data essential for the analysis task.

Attribute subset selection − In this method, where irrelevant, weakly relevant, or redundant attributes or dimensions can be discovered and deleted. Data sets for analysis can include hundreds of attributes, some of which can be irrelevant to the mining task or redundant. For instance, if the task is to arrange customers as to whether or not they are likely to purchase a popular new CD at All Electronics when notified of a sale, attributes such as the customer's telephone number are likely to be irrelevant, unlike attributes such as age or music taste.

Dimensionality reduction − Encoding mechanisms are used to reduce the data set size. In dimensionality reduction, data encoding or transformations are applied to obtain a reduced or “compressed” representation of the original data. If the original data can be reconstructed from the compressed data without any loss of information, the data reduction is called lossless.

Numerosity reduction − The data are restored or predicted by alternative, smaller data representations including parametric models (which are required to save only the model parameters rather than the actual data) or nonparametric methods including clustering, sampling, and the use of histograms.

Discretization and concept hierarchy generation − In this method, where raw data values for attributes are replaced by ranges or higher conceptual levels. Data discretization is a form of numerosity reduction that is very beneficial for the automatic production of concept hierarchies. Discretization and concept hierarchy generation are dynamic tools for data mining, in that they enable the mining of data at various levels of abstraction.

What is Data Discretization?

The data discretization techniques can be used to reduce the number of values for a given continuous attribute by dividing the range of the attribute into intervals. Interval labels can be used to restore actual data values. It can be restoring multiple values of a continuous attribute with a small number of interval labels therefore decrease and simplifies the original information.

This leads to a concise, easy-to-use, knowledge-level representation of mining results. Discretization techniques can be categorized depends on how the discretization is implemented, such as whether it uses class data or which direction it proceeds (i.e., top-down vs. bottom-up). If the discretization process uses class data, then it can say it is supervised discretization. Therefore, it is unsupervised.

If the process begins by first discovering one or a few points (known as split points or cut points) to split the whole attribute range, and then continue this recursively on the resulting intervals, it is known as top-down discretization or splitting.

In bottom-up discretization or merging, it can start by considering all of the continuous values as potential split-points, removes some by merging neighbourhood values to form intervals, and then recursively applies this process to the resulting intervals. Discretization can be implemented recursively on an attribute to support a hierarchical or multi-resolution partitioning of the attribute values, referred to as a concept hierarchy.

Concept hierarchies are useful for mining at multiple levels of abstraction. A concept hierarchy for a given numerical attribute represents a discretization of the attribute. Concept hierarchies can be used to decrease the data by collecting and restoring low-level concepts (including numerical values for the attribute age) with higher-level concepts (including youth, middle-aged, or senior). Although detail is hidden by such data generalization, the generalized data can be more meaningful and simpler to execute.

This provides a consistent description of data mining results among several mining tasks, which is a common requirement. Also, mining on a reduced data set needed fewer input/output operations and is abler than mining on a higher, ungeneralised data set. Due to these advantages, discretization techniques and concept hierarchies are generally used before data mining as a pre-processing step, rather than during mining.

Several discretization methods can be used to automatically generate or dynamically refine concept hierarchies for numerical attributes. In addition, many hierarchies for categorical attributes are implicit inside the database design and can be automatically represented at the schema definition level.